

Design of Wideband Circularly Polarized Aperture-Coupled Microstrip Antennas

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Abstract—Two variations of a novel feeding technique for a wideband circularly polarized aperture-coupled microstrip antenna are described. Prototype designs for wideband linearly polarized elements are first presented, and then used for circularly polarized designs. Techniques used for design of the feed network are detailed, for both series feed and parallel feed versions. Experimental results are shown for each antenna, and results for the two designs are compared. The impedance and axial ratio bandwidths for these antennas are among the best yet achieved for microstrip antenna elements. Several design variations are also discussed.

I. INTRODUCTION

THE aperture-coupled microstrip antenna [1] has several advantages over transmission line or probe fed patch antennas. Separate substrates can be used for the feed circuit and the antenna element to isolate spurious feed radiation from the antenna element by use of a common ground plane, and to allow more space for the feed network. The input impedance is easily controlled by the size and position of the aperture, and any excess reactance caused by the coupling aperture can be removed through the use of a tuning stub. The aperture-coupled configuration also exhibits very low cross-polarization levels, making it well suited to circularly polarized antenna designs.

A common technique for producing circular polarization is to excite two orthogonal linearly polarized elements with a 90° phase difference. This method can be utilized in the aperture-coupled microstrip antenna by using either two off-center coupling apertures [2] or a crossed slot [3]. The use of two off-center apertures results in an inherent asymmetry in the antenna which produces higher cross-polarization levels, thereby diminishing the level of circular polarization purity that can be achieved. This effect is especially apparent when a thicker antenna substrate is used to increase bandwidth. In addition, the offset slot configuration limits the slot length to less than half the patch dimension, which in turn limits the substrate thickness over which impedance matching can be obtained and, hence, the maximum bandwidth. The crossed slot retains symmetry and therefore is capable of producing circularly polarized radiation with very good polarization purity. It also permits the use of slot lengths greater than half the patch width, which is critical to achieving adequate coupling when thick antenna substrates are used for wide bandwidths.

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The aperture-coupled microstrip antenna can generally be impedance matched with an aperture that is well below resonant size, thus limiting the level of back radiation to about -20 dB relative to the main lobe. But a microstrip antenna coupled in this fashion is capable of only about 5% bandwidth, owing to the fact that the small coupling aperture limits the antenna substrate thickness that may be used. By using a thick antenna substrate with a low dielectric constant, a bandwidth of 20% to 25% can be achieved [4], [5]. However, because of the thick antenna substrate, a larger slot size is needed to obtain the necessary coupling to impedance match the antenna, resulting in a higher level of back radiation. As an alternative, the required coupling can also be achieved through the use of a "dogbone" aperture [6], which can provide increased coupling compared with a rectangular slot of the same length.

In this paper the steps taken in the design of three separate wideband circularly polarized aperture-coupled microstrip antennas are discussed, and the relative advantage and disadvantages of each design are compared. Details of a series feed design and of two parallel feed designs will be presented. The series design is capable of 15% bandwidth for an axial ratio of less than 3 dB and return loss of better than -10 dB, with a gain greater than 7 dB over this range. The parallel feed designs display even better performance, exhibiting impedance and axial ratio bandwidths of 22% and 25%, respectively, and an axial ratio of better than 2 dB.

II. BRIEF DISCUSSION OF APERTURE COUPLING USING CROSSED SLOTS AND DUAL FEEDLINES

The geometry of the aperture-coupled microstrip antenna is shown in Fig. 1. Notice that the patch is square, thereby retaining the symmetry needed for good circular polarization purity. Also notice that the coupling aperture may be a single rectangular slot for linear polarization or a crossed slot for circular polarization. The single rectangular slot will be used in the preliminary design of wideband linearly polarized elements, which will serve as the basis for a circularly polarized element using a crossed slot.

The two orthogonal apertures of the crossed slot can be modeled as short-circuited transmission lines, giving rise to an equivalent transmission line circuit. Consider only one slot being fed. Then the two slotline stubs of the unfed slot appear as series inductances to the fed slotline. With a single unbalanced offset feed line, a voltage will be induced across the unfed orthogonal slot, resulting in a high level of aperture cross-coupling, which will cause amplitude and phase errors in the network and degrade the purity of circular polarization

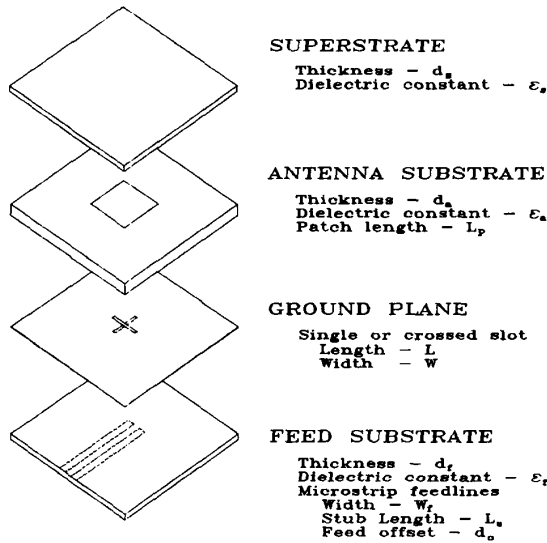


Fig. 1. Geometry of the aperture-coupled microstrip antenna. The center-to-center spacing of the microstrip feedlines is equal to $2d_o$. The stub length, L_s , is measured from the center of the aperture to the edge of the feed line.

that can be achieved. By feeding the slot in a balanced configuration with two symmetric feed lines, no voltage is induced across the unfed slot, and the two crossed apertures are perfectly decoupled. This is essential for good circular polarization purity.

Since a balanced feed is needed for the crossed slot to operate properly, the design of the linearly polarized prototype antennas should also have this balanced feed network. This consists of two feed lines offset from the center of the coupling aperture by a distance d_o , as shown in Figure 1. The addition of a second feed line to the coupling aperture causes a change in the input impedance of the antenna. An effective series impedance, Z_{eff} , is present to each feed line at the aperture when it is fed in a balanced manner. This effective impedance cannot be directly measured, but it can be easily derived from measurements by considering the dual-fed aperture-coupled microstrip antenna as a two-port network. If the S parameters of this two-port network are measured, it can be shown that the effective reflection coefficient, Γ_{eff} , seen by one feed line when both feed lines are excited in phase with equal amplitudes can be expressed as

$$\Gamma_{eff} = S_{11} + S_{12}. \quad (1)$$

Then the effective series impedance, Z_{eff} , can be found as

$$Z_{eff} = Z_0 \frac{1 + \Gamma_{eff}}{1 - \Gamma_{eff}}. \quad (2)$$

This computation can be easily performed using the trace math option on the HP8510B network analyzer.

The effective impedance may also be computed theoretically. A full-wave solution using exact Green's functions for the multilayered structure and allowing the introduction of multiple feed ports can be employed [7]. As an alternative,

a modified cavity model allowing the superstrate and multiple feed ports could be used to obtain an approximate solution [8]. A computer program utilizing the cavity model can be run much faster than the full-wave solution, making it useful for initial designs.

III. DESIGN OF THE WIDEBAND LINEARLY POLARIZED PROTOTYPE ELEMENT

A dual-fed linearly polarized element was constructed on a feed substrate of 0.031-in.-thick Rogers 5880 ($\epsilon_r = 2.2$), with an antenna substrate of 0.5-cm-thick ROHACELL foam ($\epsilon_r = 1.07$). The thick foam antenna substrate provides wide bandwidth. Fig. 2(a) shows the measured effective impedance for this element. The large bandwidth of this antenna can be explained by the fact that the aperture acts as a second resonator in combination with the patch element. This creates an effect similar to the stacked patch antenna, which uses two patches of slightly different resonant frequency to achieve a larger bandwidth. The antenna uses a large aperture with a resonant frequency that is reasonably close to that of the patch element, creating a bandwidth ($SWR < 2 : 1$) of 34% centered at 5.23 GHz. However, the fact that the resonant frequency of the aperture is close to that of the patch causes it to radiate a high level of back radiation. For this large bandwidth configuration, the level of back radiation is approximately -10 dB.

In order to lower the level of back radiation, a second element was designed on a 0.025-in.-thick Rogers 6010 feed substrate ($\epsilon_r = 10.2$). The original design of this low back lobe configuration used the same antenna substrate as in the previous case. However, the high dielectric constant of the feed substrate caused the fields to be bound tightly to the feed substrate and the aperture did not couple strongly to the patch. Therefore, the antenna substrate thickness was decreased, resulting in a reduction of bandwidth and an increase in the resonant frequency of the element. These effects can be seen in the measured effective impedance locus of Figure 2(b). Also, an aperture with a higher resonant frequency was used. This lowered the level of back radiation to -15 dB but also lowered the bandwidth. These two factors combined to lower the bandwidth to 22% centered at 5.84 GHz. Since the aperture acts as a second resonator, there is a trade-off between the bandwidth and the level of back radiation of the element; in order to increase the bandwidth the level of back radiation must also be increased, and vice versa.

A balanced feed configuration as used in the linearly polarized elements described above is difficult to implement on each arm of the crossed slot. In [3], a microstrip crossover was used in the design, but this technique creates potential fabrication problems. This problem can be overcome by implementing a 180° phase shift in the feed design, and using a series or parallel feed for the two arms of the crossed slot, as discussed below.

IV. THE SERIES FEED CONFIGURATION

The series feed configuration is shown in Fig. 3(a) and its equivalent circuit in Fig. 3(b). Each port is fed 180° out of

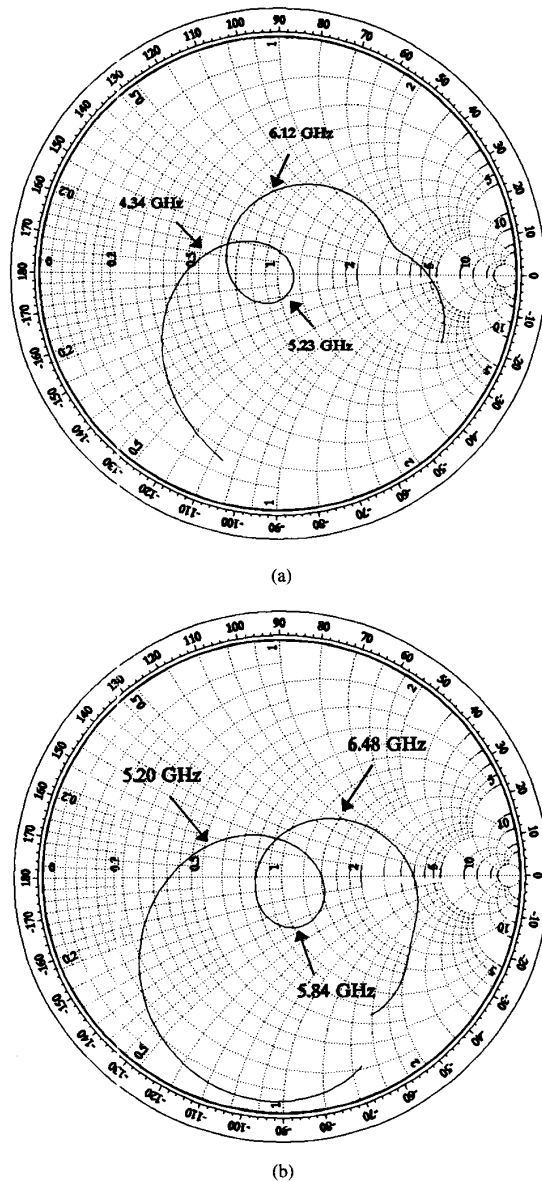


Fig. 2. (a) Effective impedance for large bandwidth linearly polarized prototype element. $d_s = 0.16$ cm, $\epsilon_s = 2.2$, $d_a = 0.5$ cm, $\epsilon_a = 1.07$, $L_p = 1.7$ cm, $L = 1.7$ cm, $W = 0.08$ cm, $d_f = 0.08$ cm, $\epsilon_f = 2.2$, $W_f = 0.232$ cm, $L_s = 0.4$ cm, $d_0 = 0.52$ cm. The frequency range is from 4 to 8 GHz. (b) Effective impedance for low back lobe linearly polarized prototype element. $d_s = 0.16$ cm, $\epsilon_s = 2.2$, $d_a = 0.32$ cm, $\epsilon_a = 1.07$, $L_p = 1.7$ cm, $L = 1.01$ cm, $W = 0.07$ cm, $d_f = 0.0635$ cm, $\epsilon_f = 10.2$, $W_f = 0.06$ cm, $L_s = 0.145$ cm, $d_0 = 0.278$ cm. The frequency range is from 4 to 8 GHz.

phase in order to account for the phase reversal caused by the oppositely directed feed lines at the two feed points for each aperture. A quarter-wavelength section of transmission line is placed between each aperture to create the 90° phase difference required for circular polarization. In order to achieve an excitation of the two orthogonal apertures that is equal in amplitude to a 90° phase difference, the effective impedance

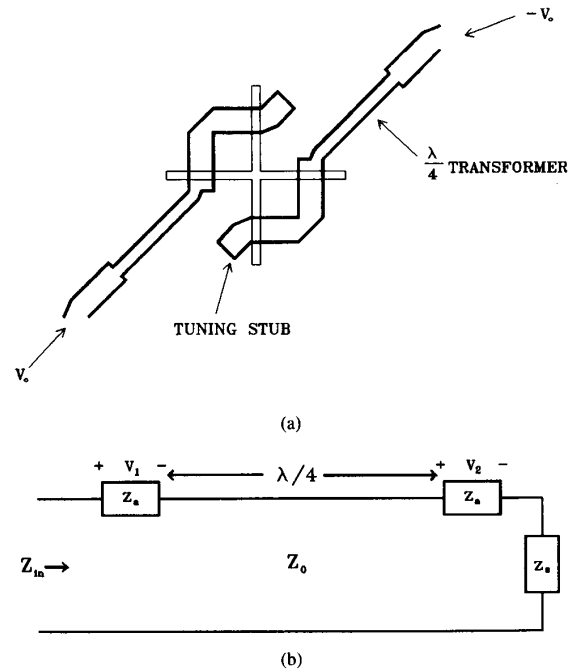


Fig. 3. (a) Series feed configuration. (b) Equivalent circuit for series feed configuration.

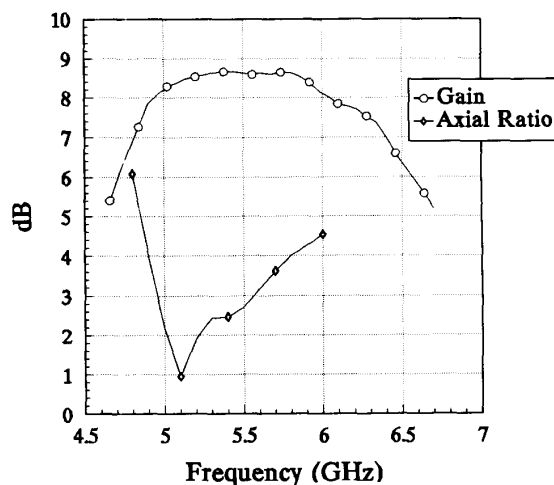
must be matched to 50Ω . Any impedance mismatch will create a standing wave on the quarter-wave section of line between the orthogonal apertures and produce phase and amplitude errors, resulting in an increased axial ratio.

In the equivalent circuit, the series impedance, Z_a , is equal to the effective impedance minus the impedance of the tuning stub. Because of the lack of a tuning stub on the first aperture, a large reactance will be present in the input impedance referenced at the first aperture. This reactance may be removed by implementing a short section of transmission line before the first aperture. The resulting input impedance can then be easily matched to 50Ω with a quarter-wave transformer.

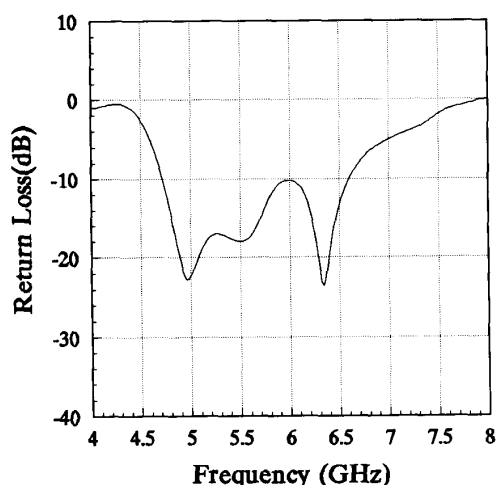
A series fed antenna was constructed from the first linearly polarized prototype design. The effective impedance, Z_{eff} , was nominally matched to 50Ω , thereby creating a VSWR of less than 2 : 1 on the quarter-wave section of line between the apertures. Since the VSWR was small a good axial ratio was obtained.

The measured gain and axial ratio of the series fed element are plotted versus frequency in Fig. 4(a), and the return loss is plotted in Fig. 4(b). Also, a measured spinning linear far-field pattern for this element is shown in Fig. 5. The axial ratio at 5.1 GHz is less than 1 dB over a wide angle; the size of the ground plane in this measurement was 61×45.75 cm. An axial ratio of less than 3 dB was obtained over a 12% bandwidth. The bandwidth for 10 dB return loss was in excess of 30%, leaving the axial ratio as the limiting factor.

It should be noted that the series feed configuration is extremely sensitive to the length of transmission line present between the two orthogonal apertures, and also to the effective impedance seen at each aperture. Therefore, small errors in



(a)



(b)

Fig. 4. (a) Axial ratio and gain plot versus frequency for large bandwidth series fed antenna. (b) Return loss plot versus frequency for large bandwidth series fed antenna.

fabrication, especially errors in alignment between the crossed slot and feed network, will cause the element to not operate properly.

V. THE PARALLEL FEED CONFIGURATION

The series feed configuration has an advantage in that it is easy to design and fabricate, but it also has several drawbacks. One disadvantage is that the feed offset from the center of the aperture must be at least $\lambda/8$ in order to accommodate a $\lambda/4$ section of transmission line between the two arms of the crossed slot. This places a severe restriction on the design of the feed network and the antenna element. Also, the axial ratio degrades fairly rapidly with frequency because of increased amplitude error in the excitation of the two apertures. This amplitude error has a much greater effect on the axial



Fig. 5. Spinning linear pattern for series fed element taken at 5.1 GHz.

ratio than does the phase error [9]. A parallel feed structure (shown in Fig. 6(a) along with its equivalent circuit in Fig. 6(b) overcomes these disadvantages. Here the two orthogonal apertures are fed through a power divider, with one arm of the output feedlines a quarter-wavelength longer than the other in order to produce a 90° phase shift.

The parallel feed structure allows for a variable feed offset, the only restriction being the separation of the feed lines near the crossed slot. Using the generally accepted microwave design procedures, this distance should be at least two substrate thicknesses from edge to edge to avoid coupling between the two arms of the parallel feed. A variable offset allows the feed points of each aperture to be moved closer to the center, thereby producing a greater amount of coupling to the antenna. This increased coupling allows a shorter aperture to be used in the design with the beneficial effect of a decreased back lobe level.

The original parallel feed configuration employed a reactive divider to feed the two slots. However, since the reactive divider does not provide any isolation between the two output ports, reflected power from one aperture can couple through to the other aperture. This coupling, albeit small, can cause enough phase and amplitude error to degrade the axial ratio beyond an acceptable level. This problem was alleviated by replacing the reactive power dividers with Wilkinson power dividers, which provide isolation between the output ports. With the two output ports isolated, the amplitude error between the two slot excitations is zero because of symmetry. An increased axial ratio is then due only to phase error.

Two antennas were constructed using the parallel feed, one using the large bandwidth configuration and the other using the low back lobe configuration of Section III. The higher dielectric constant of the feed substrate in the second element allowed thinner feed lines to be used, which in turn allowed a smaller feed offset from the center of the slots. A spinning linear pattern for the low back lobe design is shown in Fig. 7. The pattern for the large bandwidth design is the same as

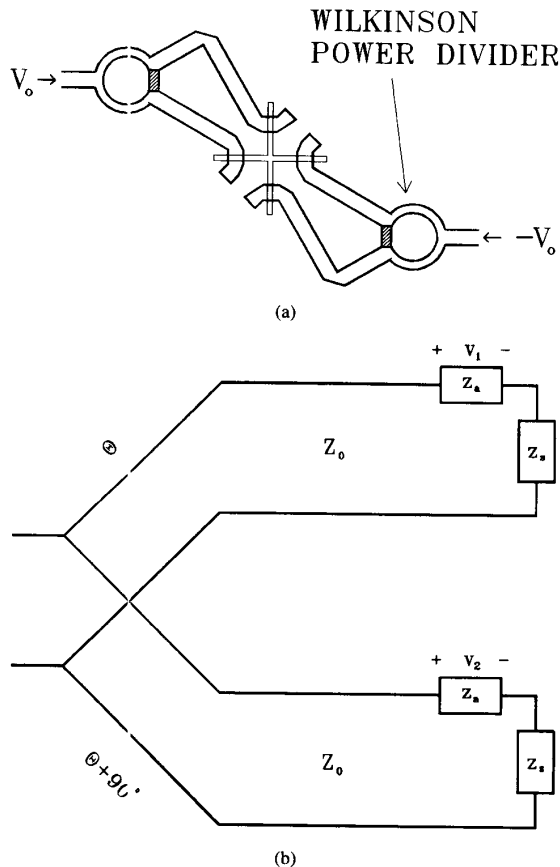


Fig. 6. (a) Parallel feed configuration. (b) Equivalent circuit for parallel feed configuration.

that for the series fed antenna, since the same linearly polarized element was used in the design. Notice again that the axial ratio is quite good over a wide angle. Parts (a) and (b) of Fig. 8 show the axial ratio and gain of each parallel feed design versus frequency. Note that the axial ratio characteristics versus frequency are much improved over the series feed, owing to the fact that amplitude error between the two orthogonal polarizations has been eliminated. The gain of the antenna then becomes the limiting factor in the bandwidth. The bandwidths for the gain, defined as being within 1 dB of the maximum gain over the band, are 29% and 22% for the larger bandwidth and low back lobe designs, respectively. As mentioned in Section III, the second antenna element was designed to have a smaller back lobe by sacrificing some bandwidth, which is evident by comparing Figs. 5 and 7, and also parts (a) and (b) of Fig. 8.

The return loss measurement for the large bandwidth parallel design (Fig. 9) shows an excellent match over a very wide band, more than 50%. This is due to the fact that some of the reflected power is absorbed in the chip resistor used in the Wilkinson power divider. The actual bandwidth for return loss less than -10 dB is the same as that of the linearly polarized element used in the design. The return loss measurement for the low back lobe parallel design is similar to Fig. 9, and is not included here.

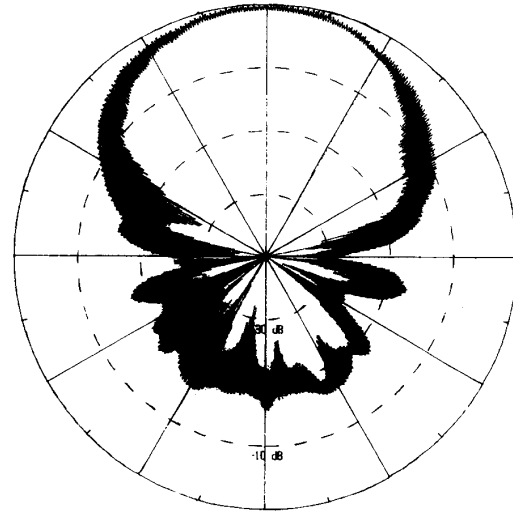


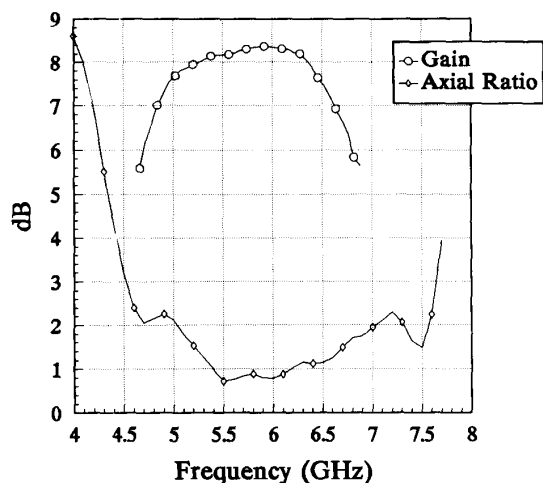
Fig. 7. Spinning linear pattern for low back lobe parallel fed antenna taken at 5.84 GHz.

VI. DISCUSSION

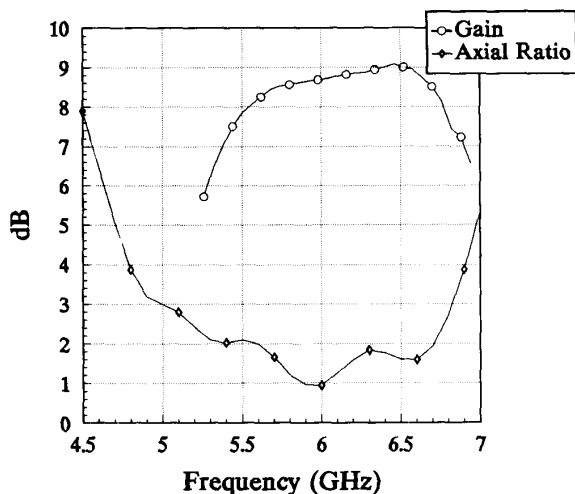
Other interesting possibilities exist for the parallel feed configuration. The Wilkinson power divider may be replaced by a quadrature hybrid, which still provides isolation between the two output lines. With the quadrature hybrid a 90° phase difference is already present at the output, thereby removing the need for an extra quarter-wavelength section of transmission line in the feed network. The quadrature hybrid will also create less phase error in the excitation of the two orthogonal polarizations over a wider frequency range, thereby establishing better axial ratio characteristics versus frequency. It should be noted here, however, that the hybrid will take up a large amount of space on the feed substrate. This could be a drawback if there are spatial limitations, such as in the design of an array. One way to reduce the size of the feed network is to switch to a higher dielectric constant for the feed substrate; however, this is not always feasible.

For a simple uniform linear or planar array, a single 180° hybrid can be used for the input to an entire feed network for several elements. For a phased array, however, each element must be fed separately. This would require a ring hybrid for each element in the array, which most likely would not be feasible because of space limitations. As an alternative in this case, a half-wavelength section of transmission line can be used to produce the 180° phase shift needed to drive both ports.

The half-wavelength section of transmission line will produce a 180° phase difference only at the center frequency, and a phase error in the excitation of each aperture will result over the band. Following the discussion presented in Section II, a phase error in the excitation of each aperture excites a mode in the orthogonal aperture. Owing to the symmetry of the crossed slot, however, the effects of this excitation on the axial ratio will cancel and no decrease in circular polarization purity will result. The input impedance match will degrade, however, as



(a)



(b)

Fig. 8. (a) Axial ratio and gain plot versus frequency for large bandwidth parallel fed antenna. (b) Axial ratio and gain plot versus frequency for low back lobe parallel fed antenna.

can be easily seen by modifying (1), which becomes

$$\Gamma_{\text{eff}} = S_{11} + S_{12}e^{j\theta} \quad (3)$$

where θ is the phase error in the excitation of the two ports. Equation (3) then can be inserted into (2) to find the effective impedance. Experimental results performed on the wide bandwidth parallel fed CP antenna presented in Section V have shown that a phase error of as much as 25° produced no noticeable effects on the axial ratio, pattern, or gain of the element over the entire bandwidth, thereby showing that a half-wavelength section of transmission line used between the two ports can operate over a wide bandwidth. This type of arrangement will not work on the series feed configuration, however, since the feed network lacks the symmetry of the parallel configuration.

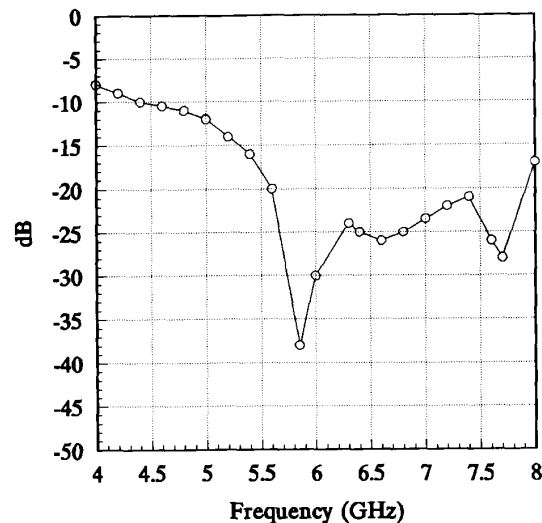


Fig. 9. Return loss for large bandwidth parallel fed antenna. Frequency range is 4 to 8 GHz. Vertical scale is 10 dB per division.

Two novel geometries for feeding a wideband circularly polarized aperture-coupled microstrip element have been presented, using either a series or a parallel feed. The parallel feed configuration provides better axial ratio and bandwidth characteristics, but is more complicated to design and fabricate because of the Wilkinson power dividers needed for isolation between the output ports. If a very large array of these elements is being constructed and a bandwidth in excess of 12% is not needed, the series feed design will greatly improve the ease of fabrication.

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